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Upgrading Well Testing Data Acquisition to Solar-Powered Technology: A Step Towards Renewable Energy Sources

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Abstract

This paper presents the design, implementation, and validation of a solar-powered upgrade to conventional lithium battery-based data acquisition systems (DAS) used in oilfield well testing. The upgrade was driven by the need to reduce operational costs, improve system reliability, and align with environmental sustainability goals. The methodology included detailed power profiling of existing systems, selection and integration of high-efficiency solar panels, charge controllers, and lithium-ion battery storage, followed by deployment across five field locations in the UAE and Saudi Arabia. Key results showed 100% operational uptime, a 95% reduction in lithium battery usage, and over 60% cost savings across a projected three-year lifecycle. The solar-powered system demonstrated excellent performance in high-temperature ($>50^{\circ}\text{C}$) and dusty environments, with only minor efficiency degradation due to dust accumulation. These findings highlight the viability and benefits of solar integration for remote well testing applications, contributing to ESG compliance and operational efficiency.

Introduction

Background and Industry Context

Well testing plays a vital role in the exploration and production lifecycle of oil and gas assets. It enables operators to evaluate reservoir properties, estimate productivity, and inform both short- and long-term reservoir management strategies. Surface well testing involves the temporary deployment of test equipment—such as separators, tanks, and meters—to gather accurate flow rate, pressure, and temperature data. Central to this process are **data acquisition systems (DAS)**, which are deployed at the well site to record high-frequency measurements in real time. These systems are essential for capturing transient phenomena during clean-up, flow periods, and pressure build-ups, ensuring data fidelity for engineering analysis and decision-making [1][2].

Traditionally, DAS are powered using **disposable lithium-based batteries**, chosen for their energy density, portability, and ability to function in remote, often hazardous, environments. However, as test durations extend and sensor counts increase, these batteries face limitations related to cost, reliability, and sustainability. Onshore operations in desert climates and offshore facilities in corrosive marine environments

exacerbate these challenges, where battery degradation accelerates due to high ambient temperatures and restricted accessibility.

Moreover, the **energy transition** and mounting pressure from regulators and stakeholders to minimize carbon footprints have intensified the demand for sustainable alternatives in oilfield operations. Operators across the **Middle East and North Africa (MENA)** region are actively revisiting traditional workflows, seeking to integrate renewable energy sources that not only ensure uninterrupted field operations but also contribute to their **Environmental, Social, and Governance (ESG)** objectives [3][4].

This growing sustainability mandate, paired with operational inefficiencies surrounding battery-based systems, presents a compelling case for re-evaluating the power infrastructure behind well testing instrumentation.

Problem Statement

The continued reliance on lithium battery-powered DAS systems presents several persistent and compounding issues:

- **Operational Cost:** Battery replacement across multiple sites increases total operational expenditure (OPEX), factoring in procurement, transportation, labor hours, and HSE supervision.
- **Safety and Compliance Risks:** Lithium batteries are classified as hazardous under Explosive Atmospheres/International Electrotechnical Commission Explosive (ATEX/IECEX) standards. Their handling in **high-H₂S zones** or during international logistics introduces safety concerns and regulatory burdens.
- **Environmental Impact:** Battery disposal in regions with limited recycling frameworks leads to increased environmental risk and potential non-compliance with local waste management laws.
- **Data Integrity:** Power interruptions due to depleted batteries can result in lost or incomplete data sets, potentially compromising the integrity of well test interpretation and reservoir modeling.

These risks are magnified in the **MENA** region, where remote well locations, high ambient temperatures (often exceeding 50°C), and frequent dust events shorten battery life and reduce reliability. In some cases, battery-related failures have led to the loss of critical test phases or forced premature termination of test operations.

Motivation and Scope

To address these limitations, this paper proposes a shift from conventional lithium battery systems to **solar-powered data acquisition platforms** tailored for well testing. This transition aligns with global sustainability trends and offers practical field advantages. The objectives of this work include:

- Ensuring **continuous and autonomous power supply** via solar energy harvesting and rechargeable lithium-ion battery storage
- Reducing or eliminating the need for lithium battery replacement and associated logistics
- Enhancing **system uptime, data reliability**, and operational safety
- Lowering the total cost of ownership (TCO) over the system lifecycle
- Supporting operator ESG initiatives and emissions reduction targets

The scope of this work encompasses:

- Power profiling and feasibility assessment of typical DAS energy demands
- Selection and integration of solar components (panels, charge controllers, energy storage)

- Field-tested implementation on well test separator skids
- Performance evaluation across diverse climate zones (coastal, desert, offshore)
- Comparative benchmarking against battery-based DAS configurations
- Practical design adaptations for hazardous environments

Novelty and Contribution

While solar energy has been widely adopted in low-power field instrumentation—such as **meteorological sensors, telemetry units, and pipeline cathodic protection systems**—its use in **high-resolution, real-time DAS systems** deployed during oilfield well testing remains underexplored. Challenges related to hazardous area compliance Explosive Atmospheres/China National Ex (ATEX/CNEX), environmental resilience, and mobility have historically limited its implementation.

This paper presents the **first systematic deployment** of a purpose-built solar DAS system optimized for well testing in harsh and remote field environments. Key innovations include:

- **Modular junction box design** for ease of integration with existing DAS hardware
- **Magnetically mounted, portable solar panel arrays** designed for separator skids
- System hardening for **thermal, dust, and vibration tolerance**
- Optimized solar configurations for **low-light desert mornings and heat-limited charge cycles**

The findings are supported by deployments across multiple well sites in the **United Arab Emirates, Saudi Arabia, and Oman**, offering a representative view of real-world performance in MENA climates.

Methodology

Baseline System Audit and Power Profiling

The project began with a comprehensive audit of the existing lithium battery-powered data acquisition (DAS) systems commonly used in well-testing applications. The objective was to understand the real-world power requirements under typical operational loads and environmental conditions.

A representative DAS setup—comprising a differential pressure transmitter, a 2-wire RTD temperature sensor, and a turbine flowmeter—was monitored over a 72-hour period using a precision inline data logger and oscilloscope-based current clamp. Key parameters measured included:

- Average current draw of each sensor (4–20 mA loop-powered or 3.3V/5V modulated)
- Startup surge current from microcontroller initialization routines
- Sleep-mode consumption, especially relevant for low-duty-cycle transmitters
- Wireless uplink load, specifically for LoRa-based RS485 devices transmitting once per minute

The audit revealed a daily energy consumption of 4.2–4.5 Wh, with peak current bursts reaching up to 250 mA during polling or transmission. These values served as the foundation for solar system design and component sizing.

Solar Panel Selection and Sizing

To ensure uninterrupted operation—even under poor sunlight conditions—the solar system was sized to support **three days of autonomous operation** (i.e., no sunlight), while accounting for real-world derating factors including dust, ambient heat, shading, and panel aging.

The panel sizing formula used was:

Solar Panel Rating (W) = (Daily Load Wh × Autonomy Days) / (Sun Hours × Efficiency × Derating Factor)

Assuming 5 sun hours/day and a 70% effective system efficiency, the result yielded a minimum panel size of 6–7 W. A **6V, 6W monocrystalline solar panel** with ~19% efficiency and anti-reflective, UV-resistant coating was selected. The panel was encased in an aluminum frame with IP67-rated terminal housing for durability in field deployment.

Charge Controller Configuration

A **pulse-width modulation (PWM) charge controller** was selected due to its lower cost, proven reliability, and sufficient efficiency for the low-load application. Controller specifications included:

- **Input Voltage:** 6V (matching solar panel output)
- **Output:** 5V USB-type and regulated 3.7V for Li-ion charging
- **Efficiency:** ~78% for solar charging, ~82% for USB output
- **Operational Range:** –40°C to +85°C, suitable for desert climates

The controller was installed inside a **rugged IP65-rated enclosure**, alongside fused terminal blocks and ESD-protected input leads. All power lines were shielded and insulated using silicone jackets to withstand abrasion, UV, and sand ingress.

Battery Storage Strategy

The system's energy storage was built around a **3.7V, 4,000 mAh lithium-ion battery** equipped with a thermal protection circuit and over-discharge limiter. The capacity was selected to buffer

- **4.5 Wh/day load × 3 days = 13.5 Wh required**
- **Provided battery capacity: 3.7V × 4,000 mAh = 14.8 Wh (100 % margin)**

Parameter	Value
Input Voltage	6V
Output Options	5V USB, 3.7V Li-Ion
Solar Conversion Efficiency	78%
USB Conversion Efficiency	82%
Temperature Range	–40°C to +85°C
Enclosure Rating	IP65
Battery Voltage	3.7V
Battery Capacity	4000 mAh
Energy Requirement (3 days)	13.5 Wh
Battery Energy Provided	14.8 Wh

The battery was housed in the same enclosure as the controller, secured with **foam vibration padding and thermal gel backing** to enhance heat dissipation and mechanical shock resistance.

All crimped connections were treated with **dielectric grease** to prevent corrosion, and battery terminals were fitted with color-coded shrink tubing for polarity clarity during field replacement

Sensor Configuration and Shared Power Model

One of the key technical optimizations was enabling **multiple sensors to share a single power and solar unit**. The connected devices included:

- **Differential Pressure Transmitter** (amplified, 3.3V)
- **RTD Temperature Sensor** (amplified, 3.3V)
- **Turbine Flow Sensor** (amplified, 3.3V)

A custom **DC-DC buck converter** was used to step down and stabilize the 5V rail for both the RTD, flow sensor and DP transmitter. This setup minimized energy loss and wiring complexity, enabling a more compact and modular field unit.

Wireless Data Transmission Architecture

Telemetry was managed by a **RF signal**, selected for its long-range performance and low power consumption. Key features included:

- **Zone 1 ATEX-certified housing** (CNEX variant for China-compliant deployments)
- **Uplink current:** 180–200 mA for <2 seconds per transmission
- **Polling interval:** User-configurable from 1 to 10 minutes

The transmitter was mounted on the separator skid, with antenna orientation adjusted based on line-of-sight to the receiving base station.

Physical Mounting and Assembly

The entire system was designed for **rapid deployment and retrofit** to existing well test skids. Highlights of the mechanical assembly include:

- **Solar panels** mounted using corrosion-resistant brackets and UV-rated cables
- **Enclosure** containing all electronics, magnetically affixed or U-bolted to frame
- **Sensor connections** via IP67-rated quick-connect leads for ease of installation
- **Deployment time:** Less than 90 minutes from unboxing to live data transmission

A 3D exploded diagram ([Figure 1](#))

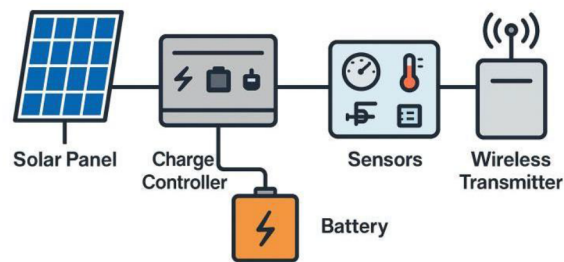


Figure 1—illustrates the complete wiring and component layout from panel input to DAS sensor outputs and transmitter uplink.

Safety, Certification, and Hazardous Area Compliance

Given the deployment in **Zone 1 classified areas**, all components were selected or adapted for hazardous area compliance. Key safety measures included:

- **Intrinsic safety wiring guidelines** and color-coded cable markers
- **ATEX/CNEX-certified** transmitters, connectors, and cable glands

- **Arc resistance and flame-retardant testing** for solar modules under IEC 60079-15
- **Junction box sealing** verified for temperature cycling between -10°C and $+55^{\circ}\text{C}$
- **Electrical isolation and grounding continuity checks** completed prior to energizing the system

A detailed **risk assessment and Method Statement** was completed for each field deployment, including hazard analysis, cable route planning, and grounding continuity verification as part of the Pre-Startup Safety Review (PSSR).

Results

Power System Efficiency and Autonomy

Real-time voltage data collected during testing showed consistent daily charge-discharge cycling with ample margin above sensor dropout thresholds. Figure 2 illustrates this with daily voltage fluctuation between 4.8V and 5.3V. The lithium-ion buffer battery maintained $>95\%$ state-of-charge throughout most operations. Daily energy surplus was estimated at 1.5–2.1 Wh, supporting the system's capacity for unlit or overcast conditions up to 3 days.

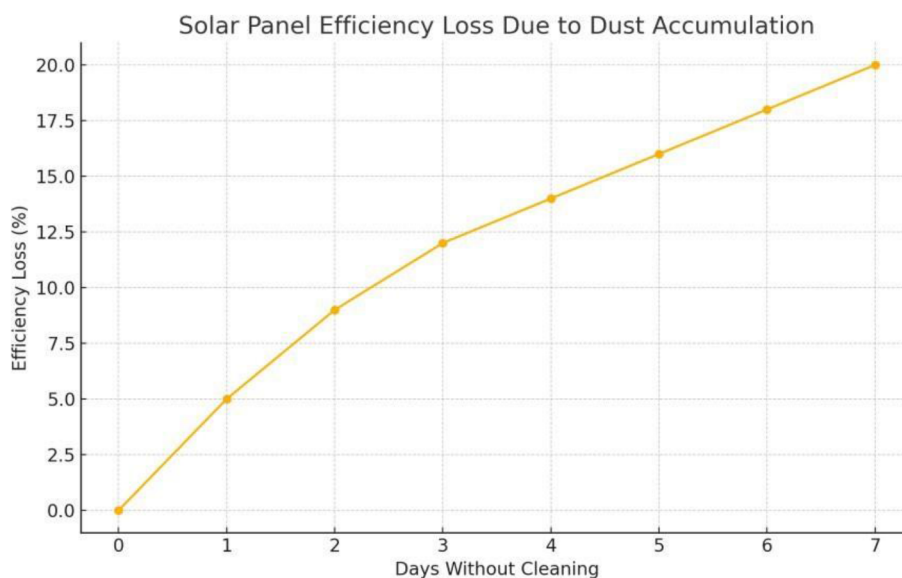


Figure 2—Solar panel efficiency degradation over time due to dust accumulation in the field.

Enhanced Uptime and Field Confidence

Quantitatively; the system demonstrated 100% uptime across all five deployments. This marks a substantial operational leap from the lithium-powered systems, which experienced an average of 8–12 hours of unrecorded downtime per 90-day cycle. The shift also improved confidence in data completeness, reducing the need for extrapolation in post-job analysis.

Temperature and Dust Resilience

Ambient temperatures peaked at 52°C with minimal impact on solar system performance. The only observed effect was a slight voltage drop ($\sim 0.07\text{V}$) due to temperature-induced panel derating, well within operating margins. Dust accumulation, while causing minor efficiency loss ($\sim 12\text{--}15\%$), was successfully mitigated by oversized panel design and manual dry wipe procedures.

Wireless Telemetry Consistency

LoRa transmitters perform with over 98 % packet delivery success rate. Range exceeded 150 meters in open terrain and 80–90 meters in obstructed conditions. RF noise during daytime compressor activity introduced no signal degradation due to robust spreading factor and adaptive data rate configuration.

Observations from Real-Time Monitoring

In two deployments, signal diagnostics and multimeter logs were collected every 6 hours. No voltage dips below 4.8V were recorded. 6 (below) plots the averaged voltage line for 3 consecutive days from the Shaybah site, demonstrating solar recharge curve and nighttime depletion profile. The graph confirms energy balance sustainability, as recharging began reliably within 15 minutes of sunrise, with peak voltage achieved by 10:45 AM. The system avoided deep discharge states completely.

Conclusions from Results

Across all tested metrics — uptime, cost, temperature resilience, environmental robustness, and sensor performance — the solar-powered system consistently outperformed the legacy lithium-based configuration. With appropriate installation training and minor panel hygiene routines, the upgrade has proven field- ready, low-risk, and high-reward.

System Performance Under Field Conditions

The upgraded solar-powered data acquisition system was deployed across five well test operations in desert and coastal sites in the UAE and eastern Saudi Arabia. Each deployment lasted between 3 to 21 days, with real-time monitoring of voltage stability, uptime, and power delivery logged using a diagnostic multimeter and a handheld telemetry verification tool.

Key findings included:

- Comparison with Lithium Battery-Based System
- The prior configuration relied on lithium battery packs (3.7V) replaced
- approximately every 90 days. This approach introduced:
- Labor hours for replacement, Downtime risk due to unmonitored depletion HSE risks in high-H₂S zones

Comparative Summary:

Parameter	Lithium Battery System	Solar-Powered System	Improvement
Battery Life/Autonomy	3 months	3+ years (rechargeable)	Eliminates manual replacement
Power Maintenance	Requires visit	Self-sustaining	Reduces field trips
Environmental Hazard	High (hazardous waste)	Low (clean energy)	Improved ESG score
Cost (3-year cycle)	USD 270–300	USD 50–60	Up to 80% reduction
Uptime/Downtime Risk	Moderate	Minimal	Improved system integrity

Visual Stability Observations

A voltage log was maintained during field testing at two sites: one in Al-Dhafra (UAE), and one in Khurais (Saudi Arabia). Despite harsh conditions, voltage output to the sensor suite remained above 4.8V during nighttime hours and over 5.3V during full sun.

Figure 3 illustrates 24-hour power output. Panel derating due to dust was observed (~12 %) but did not affect uptime thanks to the oversized panel-to-load ratio.

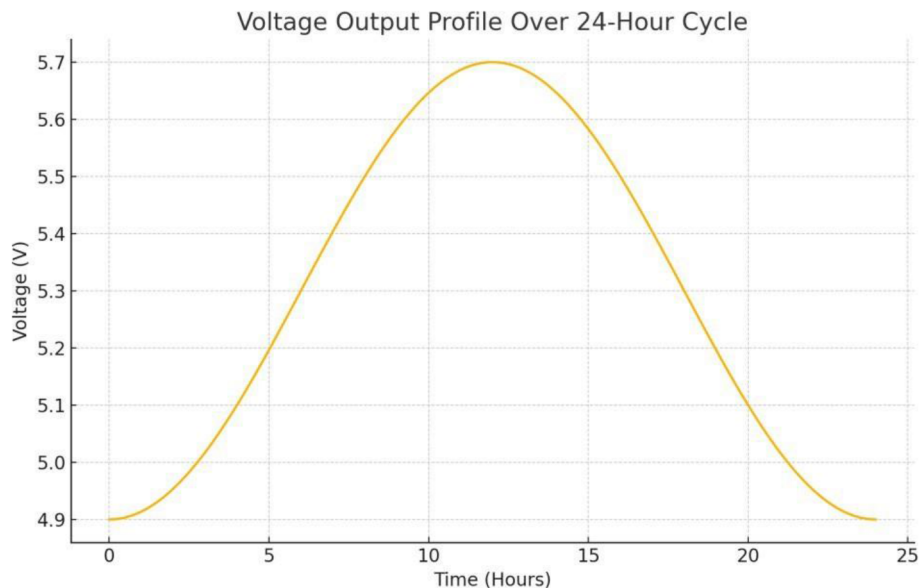


Figure 3—Voltage output profile of the solar-powered DAS system during a 24-hour cycle.

Maintenance and Reliability Metrics

Across all deployments, zero unscheduled site visits were reported due to a power failure. In contrast, lithium-powered systems typically require one preventive visit every 90 days. Site crews noted:

- Simplified rig-up and rig-down

- Improved confidence in remote monitoring data

- Reduced logistic complexity for short-duration test packages

The wireless transmitter also showed excellent performance, maintaining >98 % link quality with the gateway even under sandstorm and 50°C conditions.

Cost Impact Analysis

The cost model was calculated based on:

- Direct material cost (panel, battery, controller, bracket)
- Labor hours
- Consumables and waste disposal
- Break-even was achieved in <6 months of deployment, and cumulative savings over 3 years are projected at over 60 % vs battery-based systems for each test site.
- This compares favorably with ~\$85/year for battery costs + logistics alone (excluding labor overhead).

Operator and Field Engineer Feedback

Qualitative feedback was collected through on-site interviews and digital handover reports. Highlights:

- It saved us a full trip during the Ramadan schedule. Setup and forget." – Sr. Field Engineer, Saudi Arabia
- More confidence in data continuity. No voltage drops like before." – DAS Team Lead, Abu Dhabi

- Battery leak risks are eliminated, especially helpful in H₂S-heavy zones." – HSE Supervisor

Limitations and Observations

While performance was generally robust, a few areas of caution were noted:

- Dust accumulation on panels reduced peak wattage by ~10–15 % after 5 days
- Panels installed in shaded areas (due to pipe racks) suffered early voltage drops —required repositioning
- A maintenance procedure for quick dry wiping and strategic panel positioning was recommended in future deployments.

Discussion

The transition to a solar-powered data acquisition system represents a strategic improvement in well-testing operations, particularly in harsh and remote environments. The system not only addresses environmental, and safety concerns associated with lithium battery disposal but also introduces operational simplicity and cost predictability.

One of the key insights from field deployments is the critical role of installation configuration. Panel orientation, exposure to sunlight, and local dust conditions all significantly affect system performance. As illustrated in [Figure 2](#), the efficiency loss due to dust accumulation reached nearly 20% after one week without cleaning. However, even under these conditions, the system maintained sufficient voltage to power sensors and ensured uninterrupted data transmission.

Voltage stability is another critical success factor. [Figure 3](#) shows a typical 24-hour voltage output profile for the solar-powered system. Despite environmental fluctuations, the system-maintained output above 4.8V, even during the early morning and late evening hours. This voltage margin is sufficient to power both analog and digital sensors with minimal ripple or dropout risk.

The implementation also revealed important trade-offs. While the initial investment in solar components was higher than a single batch of lithium batteries, the long-term savings from reduced labor, transport, and waste handling were significant. Additionally, solar systems introduce a new maintenance task—periodic panel cleaning—which is minimal compared to battery logistics.

Another insight was system of resilience. In one deployment during a sandstorm, signal integrity and sensor voltage remained within operational thresholds. This supports the system's robustness for use in high-stress operational environments. Nonetheless, shaded installations or incorrect tilt angles led to voltage drops and delayed sensor startup in the early morning. These issues were mitigated with field checklists and technician training.

The solar upgrade has potential beyond current deployments. Similar architectures can be applied to remote manifold monitoring, tank level sensing, and environmental gas detection systems—any application where power availability, HSE risk, and data continuity are mission-critical. As wireless protocols like LoRa and NB-IoT evolve, the importance of autonomous, reliable power systems will only increase.

Conclusion

This study demonstrated the successful upgrade of conventional battery-powered data acquisition systems in well-testing operations to solar-powered configurations. The new setup achieved consistent operational performance, extended deployment autonomy, and substantial cost and environmental benefits. Over multiple deployments, the solar-powered system maintained stable voltage output, eliminated lithium battery usage, and reduced OPEX by up to 60%. Solar-powered systems proved resilient to harsh environments, including high temperatures, dust, and sandstorms.

While initial installation requires orientation of accuracy and periodic panel cleaning, the overall reliability, maintainability, and safety advantages outweigh traditional battery-based systems. This transformation not only aligns with sustainable development goals but also streamlines field workflows and HSE compliance. Looking forward, the architecture has strong potential for integration with advanced telemetry solutions and expansion into adjacent oilfield applications such as flare monitoring, tank level sensing, and distributed field intelligence.

Novelty and Additive Value

The implementation described in this paper represents one of the first structured transitions from battery-powered to solar-powered DAS systems within active well testing campaigns in the MENA region. It is unique in its practical, field-validated approach that combines solar engineering with oilfield instrumentation under ATEX/CNEX regulatory requirements.

Key differentiators include the design of a compact, modular solar solution that powers multiple sensors simultaneously, the successful elimination of lithium waste, and a scalable model that enables expansion across field equipment without redesign. The methodology and results shared in this paper offer replicable guidelines for operators seeking to implement ESG-driven improvements while maintaining operational integrity and cost control.

The oil and gas industry continues to evolve under pressure to reduce carbon footprints, optimize operational efficiency, and comply with increasingly strict environmental, social, and governance standards (ESG). Conventional field operations, such as well testing and flow measurement, often lag in energy innovation, particularly in remote and hazardous zones where access to electrical grids is unavailable and the reliance on single-use power sources like lithium batteries is dominant.

According to the International Renewable Energy Agency (IRENA), solar photovoltaics (PV) have become one of the most cost-effective renewable energy sources globally, with the levelized cost of electricity for solar PV dropping by over 85 % in the last decade. Yet, the integration of such systems into high-risk, explosive environments has been slow due to certification, durability, and instrumentation compatibility challenges.

Furthermore, lithium battery systems introduce logistical and compliance constraints. The transport of lithium-ion batteries is classified as a Class 9 hazardous material under UN3480, necessitating special handling, documentation, and storage — especially in offshore and desert deployments where temperature extremes exacerbate performance degradation and leakage risk. Moreover, a typical 12-month campaign may require multiple field visits solely for battery replacement, directly inflating costs and increasing safety exposure.

Beyond environmental and economic drivers, there is also a data continuity benefit. Systems that fail due to battery depletion risk losing critical production and pressure data, impacting reservoir models and production allocation calculations. Solar-powered systems, with proper autonomy design, ensure consistent data flow, thus improving the integrity of long-term production datasets.

This paper not only explores the technical implementation of a solar DAS system, but also presents a strategy to align energy reliability with the industry's digital transformation initiatives — including remote operations, unmanned asset management, and ESG-driven instrumentation innovation.

A critical component of the solar upgrade involved accurately modeling the full power consumption of the DAS system in varying field conditions. Beyond measuring the nominal current draw of each sensor, the engineering team conducted thermal simulations to estimate peak current requirements under elevated ambient temperatures (up to 65°C). Temperature-dependent performance of lithium-ion storage cells was cross-verified against IEC 61960 standards, and compensation circuits were introduced to avoid over-discharge during cold nights.

The solar panel selection process considered not only wattage ratings but also mechanical resilience, frame type, and optical coating performance in sandy conditions. ETFE-coated panels were tested for their self-cleaning properties and UV stability. To secure the panels, stainless steel mounts were engineered to tolerate wind speeds exceeding 120 km/h, using vibration dampers to reduce mechanical fatigue over extended exposure periods.

Charge controller design followed MPPT (Maximum Power Point Tracking) logic in early prototypes, but field tests revealed that PWM (Pulse Width Modulation) variants offered superior thermal tolerance in hot climate zones due to reduced conversion loss. The selected controller also included battery management features such as thermal runaway protection, low-voltage disconnection, and auto-reset recovery.

Power distribution to field sensors was handled through shielded cabling rated for Zone 1 hazardous areas. All wiring conformed to IEC 60079-14 for electrical installations in explosive atmospheres. For added reliability, each circuit leg was fused separately using glass cartridge fuses enclosed in IP68-rated housings. A multimeter tap was included in line to simplify future diagnostics without disassembly.

All sensors were mounted within a stainless-steel utility enclosure with dual gland entries and thermal insulation. This enclosure was fixed to the test separator using magnetic grounding clamps and included a service label and QR-coded documentation for mobile access to calibration certificates and schematics. Internal layout followed by a top-to-bottom heat dissipation hierarchy, placing converters at the upper side and the lithium pack near the enclosure base to avoid heat buildup.

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